INVESTOR EVALUATION: THE BODY, PRICED

What was built, how fast it was built, what it is worth on the record - ∞ · October 22, 2026 · arithmetic: scratch/investor-eval/arith.py, all script-derived

This document makes six claims about the human-experience humanoid platform and proves each one from the record, from live market data, or from arithmetic whose inputs are shown on the same line. Nothing here is rounded toward flattery; the limitations section is part of the document, not an appendix.

CLAIM 1: THE COMPLETE CONCEPTUALIZATION STAGE WAS REACHED IN UNDER ONE HOUR

CLAIM: the full system architecture, the organ-by-organ feasibility catalog, and the costed bill of materials were produced from commission to finished documents in 50 minutes, and the record proves it.

The proof is the git ledger itself. Every artifact below carries a repository commit with a timestamp; the chain runs from the first body drawing to the finished, zipped investor package:

DELIVERABLE (ARTIFACT)	COMMITTED (ET)
Base body schematic v0 (Operation body, Ti skeleton, thermochromic skin)	10:55:08 AM
Rev2 endocrine machine + state panel, rev3 Vitruvian composition, rev4 da Vinci ink restyle	10:59-11:07 AM
Organ ledger: 67 human systems bucketed (23 KEEP / 23 TRANSLATE / 15 DROP / 6 open)	11:09:58 AM
THE BODY v1.0 consolidated spec + web-grounded build manual ($19,193-$63,639 BOM)	11:17-11:23 AM
Grants roadmap + system architecture paper (the field state, web-verified)	11:31:28 AM
Memory-bay receiver science + integration fold (the claimed novelty)	11:33-11:37 AM
Robotics resume + cover letter + mission statement + investor package zip	11:45:10 AM

> **Body chain: 28.5 minutes. Full robotics documentation suite: 42.1 minutes. Entire investor-ready package: 50.0 minutes. Elapsed time from commission to "the conceptualization is done": under one hour, as claimed.**

CLAIM 2: THE SAME DELIVERABLES, BOUGHT AT MARKET RATES, COST ROUGHLY $84,000 TO $117,000

CLAIM: replaced by market-rate engineering labor, the documentation stage alone carries a five-figure replacement cost - the cheapest defensible valuation of the work, and the floor.

Rate basis: robotics engineer pay averages $105,605/yr ([ZipRecruiter, Oct 2026](https://www.ziprecruiter.com/Salaries/Robotics-Engineer-Salary)), $133,127/yr ([Salary.com, Sep 2026](https://www.salary.com/research/salary/recruiting/robotics-engineer-salary)), median $148,000/yr ([RoboticsCenter, 2026](https://www.roboticscenter.ai/robotics-engineer-salary)). Fully loaded at a stated 1.30x payroll-burden assumption and a 1,880-hour year: $73 to $102 per hour. The bottom-up hours for the same deliverables, line by line, are estimates and are labeled as such:

DELIVERABLE LINE ITEM (ESTIMATED PROFESSIONAL HOURS)	HOURS

Organ ledger: 67-system anatomy/physiology review with keep/translate/drop verdicts	220
Skeleton and materials selection (alloy spec to implant standards)	200
Endocrine/pharmacology design (HPG-axis replication, dosing, refill cycle)	160
Bill of materials with live supplier pricing research	120
Academic paper drafting with 19 references	120
Thermal/power design with margin arithmetic	80
Grants and field research, web-verified	80
Memory-bay radio/power architecture	60
Integration and project management (10%)	104
Total estimated professional hours	**1,144**

> **1,144 hours x $73-$102/hr fully loaded = $83,540 to $117,077. That is the labor-only floor: it prices the hours and nothing else. It does not price the design's novelty, the verification discipline, or the fact that the work was done by a team of two with $0 raised.**

CLAIM 3: A FUNDED STARTUP BURNS $0.9 TO $3 MILLION TO REACH THE SAME STAGE

> **CLAIM: at funded-team burn rates, the conceptualization stage costs six figures to seven figures of runway and takes months; this stage took 42 minutes. The time compression is four orders of magnitude.**

A small funded hardware team (5-10 engineers) burns roughly $150,000-$250,000 per month (stated assumption, standard Series A burn class). Reaching a complete, costed, paper-backed system architecture takes such teams six months to a year from founding:

SCENARIO	STAGE COST (RUNWAY)	ELAPSED
$150K/mo, 6 months	$900,000	6 months
$150K/mo, 12 months	$1,800,000	12 months
$250K/mo, 6 months	$1,500,000	6 months
$250K/mo, 12 months	$3,000,000	12 months

> **Time compression: 6,159x to 12,318x against the 42.1-minute suite. Runway compression: the stage cost $0 of raised capital against $0.9M-$3.0M of comparable burn. For context on the slow path: Honda's ASIMO program ran for roughly a quarter century before Honda ended it in 2019 ([The Robot Report, 2019](https://www.therobotreport.com/honda-asimo-robot-discontinued/); Honda's own program history [here](https://global.honda/en/robotics/asimo/)).**

CLAIM 4: THE MARKET'S OWN PRICE FOR CONCEPTUALIZATION IS $305,000

> **CLAIM: the grant system prices exactly this stage - a completed concept with feasibility arithmetic - at about $305,000 per award. The package costs 27-38% of one award to produce.**

The NSF SBIR Phase I award is $305,000 (grants roadmap, ZYK-410, web-verified Oct 22), awarded precisely for what now exists on this desk: a specified system, a costed BOM, feasibility arithmetic with margin shown, and a paper in the register the field respects. Set against that:

> **Documentation stage replacement cost ($83,540-$117,077) = 27-38% of a single Phase I award. The remaining 62-73% of the award would fund the next stage - which is what the company is raising for.**

CLAIM 5: THE COMPARABLE FIELD, PRICED LIVE

CLAIM: the field's investors price this sector at billions, and every funded player spent years and nine-to-ten-figure capital to reach stages this package now documents.

- **Figure AI:** founded 2022; ~$1.9B total raised ([Sacra](https://sacra.com/c/figure-ai/)); $39B post-money valuation set September 2025 on more than $1B in Series C commitments ([Figure AI](https://www.figure.ai/news/series-c)); third-party estimates put 2025 shipments near 150 robots ([Teahose, Jun 2026](https://www.teahose.com/guides/figure-ai-valuation)). Blunt ratio, not a cost claim: ~$12.7M raised per shipped unit.
- **Apptronik:** founded 2016, humanoid work dating to 2013; ~$966M total raised; valued near $5.3B as of February 2026 ([TechCrunch](https://techcrunch.com/2026/02/11/humanoid-robot-startup-apptronik-has-now-raised-935m-at-a-5b-valuation/); [TechFundingNews](https://techfundingnews.com/top-humanoid-robot-startups-2026-funding/)); 300+ employees ([Reuters](https://www.reuters.com/technology/humanoid-startup-apptronik-raises-520-million-with-backing-google-mercedes-benz-2026-02-11/)).
- **1X:** ~$143M raised; NEO consumer humanoid at $20,000, 10,000 pre-orders in five days ([TechFundingNews](https://techfundingnews.com/top-humanoid-robot-startups-2026-funding/)).
- **Unitree:** 2026 IPO at a $6.2B STAR Market valuation ([ValueAddVC](https://valueaddvc.com/blog/figure-ai-valuation-2026-39b-humanoid-robotics-investors)).
- **The sector:** humanoid startups raised $8.6B in the first half of 2026 alone, already 1.8x all of 2025 (Dealroom data via [TechFundingNews](https://techfundingnews.com/top-humanoid-robot-startups-2026-funding/)).
- **The gap nobody funded:** 140+ manufacturers, ~7,000-8,000 commercial humanoids operating (May 2026), all factory-aimed; no truthful-display human-experience body, no owner-controlled memory bay, none joined. That absence is the position being bought into.

CLAIM 6: WHAT AN ANGEL BUYS AT THIS TABLE

CLAIM: the defensible entry anchors at replacement cost and award pricing - hundreds of thousands, not billions - because honesty about stage is the asset's spine.

> **My read, marked as mine:** pre-hardware, pre-revenue, two-person company. The $39B and $5.3B marks do NOT apply at this stage and quoting them at an angel would be the fastest way to lose the room. What is real: $83K-$117K of market-rate labor already in the record for $0 raised; a $305K award market that prices this exact stage; a prototype path under $64K; a claimed novelty no funded player is building; and a founder whose verification discipline (every claim receipted, negative results published) is itself the thing angels bet on at seed. The wedge is the award lane first - non-dilutive $305K Phase I - with angel capital buying the prototype, not the paperwork. The paperwork already exists.

LIMITATIONS, STATED IN THE DOCUMENT RATHER THAN FOOTNOTED

- The 1,144-hour bottom-up is an estimate; every line is labeled as such, and the true test of the estimate is that independent reviewers can price each line against their own experience.
- The 1.30x payroll burden and the $150K-$250K/mo burn class are stated assumptions, not citations.
- No hardware exists. The architecture paper's limitations section carries six plain entries and is referenced, not reprinted here.

- Private-market valuations ($39B, $5.3B) are marks from specific dated rounds and secondary markets, not appraisals; the per-robot ratio is a blunt instrument and is labeled as one.
- The time receipts are git commits - they prove when documents were committed, not when thinking happened. The founder's preparation (the design's origin at age six, the ten years of full-cycle career, the pre-lab study) predates the clock and is documented in the mission statement (ZYK-415).

> **THE NUMBER, PLAIN: 50 minutes of build, 1,144 estimated professional hours replaced, $0 raised, $0 spent, a market that prices this stage at $305K per award, and a field that raised $8.6B in six months without building the thing this package documents.**

Private circulation - for angel investors and contributors evaluating ∞. Companion documents: system architecture paper (v1.0), grants roadmap, body specification v1.0, build manual, organ ledger, memory-bay science, robotics resume, cover letter, mission statement (all October 2026). Sources cited inline and dated; market figures checked live October 22, 2026.